

Formal Submission for Correction of Quantum Terminology

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This document formally submits a correction to longstanding misnomers in particle physics terminology, including but not limited to the terms "flavor" and "color." These metaphorical names have introduced ambiguity and hindered clarity in the communication and teaching of quantum chromodynamics (QCD) and related fields. To strengthen the integrity and accuracy of the discipline, this submission proposes operational replacements grounded in mathematical and physical precision.

Corrections Proposed:

1. **Flavor** → *Mass/Decay Vector*: Denotes the quantum identifier linked to a quark's charge, rest mass, and decay pathways.
2. **Color** → *Strong-Force Charge*: Represents the SU(3) charge necessary for confinement and gluon-mediated interactions.
3. **Strangeness/Charm** → *Weak-Interaction Symmetry Labels*: Operational identifiers tied to weak-force couplings and conserved currents.

Resolution of the Matter–Antimatter Asymmetry (Baryon Enigma):

The correction is accompanied by a mechanism explaining the matter-dominant universe using a chirality-driven torsional bias. The effective Lagrangian extends the Standard Model as follows:

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \frac{1}{2} (\partial\tau)^2 - \frac{1}{2} m_\tau^2 \tau^2 - \frac{\kappa}{3} \tau^3 - \frac{\lambda}{4} \tau^4 + g u_\mu J_5^\mu \tau$$

The cubic term $\kappa\tau^3$ provides a parity-odd chiral bias, introducing an axial chemical potential:

$$\mu_5(T) = g \tau_{\text{eff}}(T)$$

This leads to a baryon number evolution governed by:

$$dn_B/dt \approx \Gamma_{\text{sph}}(T) N_f/6 \mu_5(T) T^2 - \Gamma_{\text{wash}}(T) n_B$$

With realistic parameterization, the predicted baryon-to-photon ratio is:

$$\eta_B \approx 6.1 \times 10^{-11}$$

This value aligns with the Planck 2018 cosmic microwave background data (observed $\eta_B \approx 6 \times 10^{-11}$).

Empirical Falsifiability:

1. Heavy-Ion Collisions: Predicts polarization asymmetry $\Delta P_{\Lambda-\Lambda^*}$ at the 10^{-4} to 10^{-3} level.
2. Cosmic Microwave Background: Predicts TB/EB parity-odd correlators detectable by upcoming observatories.

This submission corrects flawed terminology, replaces it with operational definitions, and demonstrates a resolved and testable mechanism for the baryon asymmetry of the universe. These steps uphold the integrity of physics and ensure accurate communication for current and future generations.

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